

Tchebycheff Shadow Perspective Mapping for Computing the Integral R_2 Indicator

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Abstract

This note presents a self-contained perspective-mapping view of the continuous, integral R_2 indicator. We work in minimization form after translating the ideal point to the origin. Hence every approximation point is a strictly positive loss vector; equivalently, the original ideal point strictly dominates every point of the approximation set. The subgraph of the lower Tchebycheff shadow over the weight simplex is mapped to the complement of an anchored-box union in reciprocal objective space. This gives a direct formula for the absolute integral R_2 value. It also gives the usual anchor-normalised improvement $R_2(\{a\}) - R_2(P)$ as a finite weighted hypervolume difference with density $(x_1 + x_2 + x_3)^{-4}$ in three objectives. Consequently, a hypervolume algorithm that emits a disjoint box decomposition can be reused: ordinary box volumes are replaced by closed-form weighted box integrals. If the tree-free three-dimensional hypervolume algorithm described in [2] emits $M = n$ boxes in $O(n \log n)$ time, then the integral R_2 value is obtained in $O(n \log n)$ time and $O(n)$ space after adding one dominated anchor. The same perspective principle extends to N objectives, with density $(x_1 + \dots + x_N)^{-(N+1)}$ and $O(2^N M)$ weighted-integration overhead for an M -box decomposition. A Python reference implementation verifies the formula against a subdivision-based exact evaluator and Monte Carlo integration.

1 Introduction

The integral R_2 indicator evaluates an approximation set by integrating the best weighted Tchebycheff scalarizing value over a continuous weight domain. Each weight vector defines a scalar preference model; the lower envelope of the corresponding Tchebycheff shadows records the best scalarizing value attained by the approximation set. Exact computation is attractive because finite weight sampling can miss small regions of the weight simplex and therefore may fail to reflect fine changes of this lower envelope.

This report describes a geometric route to exact computation. The key observation is that the perspective transformation

$$x_i = \frac{w_i}{t}$$

turns the subgraph of the Tchebycheff lower envelope into a reciprocal anchored-box complement. The Jacobian of this map is not constant. Hence integral R_2 is not an ordinary hypervolume value. It is a *weighted* hypervolume-type integral. Nevertheless,

all geometric decomposition work of a hypervolume algorithm can be reused: once the reciprocal dominated region is decomposed into disjoint axis-aligned boxes, each ordinary box-volume term is replaced by a closed-form weighted box integral.

The three-objective case is special because efficient hypervolume decomposition is particularly well understood. The dimension-sweep algorithm of Fonseca, Paquete, and López-Ibáñez [1] and the tree-free array implementation described in [2] provide the geometric integration region. If such a routine emits $M = n$ boxes in $O(n \log n)$ time, the perspective-weighted integral has only linear overhead in M , and the resulting exact computation of the integral R_2 value is $O(n \log n)$ in three objectives.

The report is organized as follows. Section 2 fixes the notation. Section 3 gives a two-dimensional warm-up with an inline TikZ visualization and a hand calculation. Sections 4–6 give the three-dimensional absolute and improvement formulas and the $O(n \log n)$ consequence. Section 7 extends the construction to N objectives and provides pseudocode, correctness, and complexity statements. Section 9 summarizes the Python verification.

2 Setting and notation

We consider minimization. The ideal point is translated to the origin. The standing assumption is therefore

$$P = \{p^1, \dots, p^n\} \subset \mathbb{R}_{>0}^N.$$

Equivalently, in the original objective space, the ideal point strictly dominates every approximation point. The coordinate p_i is the positive loss of point p in objective i .

For N objectives the weight domain is

$$\Delta_{N-1} = \{w \in \mathbb{R}_{\geq 0}^N : w_1 + \dots + w_N = 1\}.$$

For a point $p \in \mathbb{R}_{>0}^N$ define its weighted Tchebycheff shadow by

$$g_p(w) = \max_{i=1,\dots,N} w_i p_i.$$

For a finite set P , the lower Tchebycheff envelope is

$$\tau_P(w) = \min_{p \in P} g_p(w),$$

and the unnormalised integral R_2 value is

$$R_2(P) = \int_{\Delta_{N-1}} \tau_P(w) \, dw.$$

If one wants the probability average over the simplex, this unnormalised value is divided by $\text{vol}_{N-1}(\Delta_{N-1}) = 1/(N-1)!$.

A point $a \in \mathbb{R}_{>0}^N$ is called a dominated anchor for P if

$$a_i \geq p_i \quad \text{for all } p \in P \text{ and all } i.$$

The anchor-normalised improvement is

$$I_a(P) = R_2(\{a\}) - R_2(P).$$

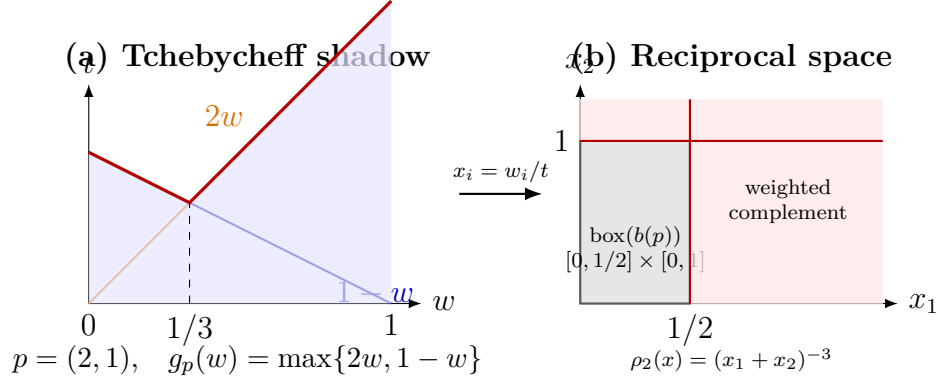
The improvement is convenient for subset selection and greedy approximation. The absolute value $R_2(P)$ is recovered exactly from the same computation by adding the single-anchor value $R_2(\{a\})$.

3 Warm-up in two objectives

The perspective idea is easiest to see in two objectives. The weight simplex is the interval $w \in [0, 1]$, where $w_1 = w$ and $w_2 = 1 - w$. For a single point $p = (p_1, p_2)$,

$$g_p(w) = \max\{wp_1, (1 - w)p_2\}.$$

The integral $R_2(\{p\})$ is the area under this V-shaped Tchebycheff shadow.



For the concrete point $p = (2, 1)$, the kink solves $2w = 1 - w$, hence $w = 1/3$. The scalarization-side computation is

$$\begin{aligned} R_2(\{p\}) &= \int_0^{1/3} (1 - w) \, dw + \int_{1/3}^1 2w \, dw \\ &= \left[w - \frac{w^2}{2} \right]_0^{1/3} + \left[w^2 \right]_{1/3}^1 = \frac{5}{18} + \frac{16}{18} = \frac{7}{6}. \end{aligned}$$

Now apply the two-dimensional perspective map

$$x_1 = \frac{w}{t}, \quad x_2 = \frac{1 - w}{t}, \quad t = \frac{1}{x_1 + x_2}.$$

The reciprocal box corner is $b(p) = (1/2, 1)$. The complement of the box is decomposed as

$$\mathbb{R}_{>0}^2 \setminus [0, 1/2] \times [0, 1] = [1/2, \infty) \times [0, \infty) \cup [0, 1/2] \times [1, \infty).$$

The Jacobian is $(x_1 + x_2)^{-3}$, so

$$\begin{aligned} R_2(\{p\}) &= \int_{1/2}^{\infty} \int_0^{\infty} \frac{dx_2 \, dx_1}{(x_1 + x_2)^3} + \int_0^{1/2} \int_1^{\infty} \frac{dx_2 \, dx_1}{(x_1 + x_2)^3} \\ &= 1 + \frac{1}{6} = \frac{7}{6}. \end{aligned}$$

This example shows the whole mechanism: the Tchebycheff-shadow subgraph becomes a weighted reciprocal-space box complement.

4 Absolute integral R_2 by perspective mapping in three objectives

We now return to three objectives. For $p \in \mathbb{R}_{>0}^3$ define the reciprocal box corner

$$b(p) = \left(\frac{1}{p_1}, \frac{1}{p_2}, \frac{1}{p_3} \right),$$

and for a set $Q \subset \mathbb{R}_{>0}^3$ define the anchored-box union

$$U(Q) = \bigcup_{q \in Q} [0, q_1] \times [0, q_2] \times [0, q_3].$$

The three-dimensional perspective map is

$$x_i = \frac{w_i}{t}, \quad w_i = \frac{x_i}{x_1 + x_2 + x_3}, \quad t = \frac{1}{x_1 + x_2 + x_3}.$$

Lemma 1 (Pointwise box correspondence). *For $p \in \mathbb{R}_{>0}^3$ and $t > 0$, apart from measure-zero boundaries,*

$$t > g_p(w) \iff x \in [0, b_1(p)] \times [0, b_2(p)] \times [0, b_3(p)].$$

Proof. Using $w_i = x_i/(x_1 + x_2 + x_3)$ and $t = 1/(x_1 + x_2 + x_3)$, the inequality $t > g_p(w)$ becomes

$$\frac{1}{x_1 + x_2 + x_3} > \max_i \frac{x_i p_i}{x_1 + x_2 + x_3}.$$

Multiplying by the positive denominator gives $1 > \max_i x_i p_i$, which is equivalent to $x_i < 1/p_i$ for all i . $\square$

Lemma 2 (Jacobian). *The perspective transformation has volume element*

$$dw \, dt = (x_1 + x_2 + x_3)^{-4} dx_1 dx_2 dx_3.$$

Proof. Use local coordinates (w_1, w_2, t) with $w_3 = 1 - w_1 - w_2$ and inverse

$$w_1 = \frac{x_1}{s}, \quad w_2 = \frac{x_2}{s}, \quad t = \frac{1}{s}, \quad s = x_1 + x_2 + x_3.$$

The absolute determinant of the Jacobian of (w_1, w_2, t) with respect to (x_1, x_2, x_3) is s^{-4} . $\square$

Theorem 1 (Absolute weighted-complement representation). *For every nonempty $P \subset \mathbb{R}_{>0}^3$,*

$$R_2(P) = \int_{\mathbb{R}_{>0}^3 \setminus U(b(P))} \frac{dx_1 dx_2 dx_3}{(x_1 + x_2 + x_3)^4}.$$

Proof. By definition,

$$R_2(P) = \int_{\Delta_2} \tau_P(w) dw = \int_{\Delta_2} \int_0^{\tau_P(w)} dt dw.$$

The condition $t < \tau_P(w)$ means $t < g_p(w)$ for every $p \in P$. By Lemma 1, this is equivalent to saying that x is outside every reciprocal anchored box, i.e. $x \notin U(b(P))$. Lemma 2 supplies the density. $\square$

5 From absolute R_2 to finite improvement regions

The absolute representation in Theorem 1 is conceptually simple, but its integration domain is unbounded. For computation it is convenient to introduce a dominated anchor a and use

$$R_2(P) = R_2(\{a\}) - I_a(P), \quad I_a(P) = R_2(\{a\}) - R_2(P).$$

The improvement region is bounded away from the singular origin and is exactly a reciprocal anchored-box difference.

Theorem 2 (Weighted hypervolume representation of improvement). *Let $P \subset \mathbb{R}_{>0}^3$ and let $a \in \mathbb{R}_{>0}^3$ be componentwise worse than every $p \in P$. Then*

$$I_a(P) = R_2(\{a\}) - R_2(P) = \int_{U(b(P)) \setminus U(b(a))} \frac{dx_1 dx_2 dx_3}{(x_1 + x_2 + x_3)^4}.$$

Proof. Subtract the absolute formula of Theorem 1 for P from the corresponding formula for the singleton $\{a\}$. Since a is componentwise worse, $b(a)$ is componentwise smaller and $U(b(a)) \subseteq U(b(P))$ whenever P contains at least one point improving upon a . The difference of the two complements is therefore $U(b(P)) \setminus U(b(a))$. $\square$

The singleton value $R_2(\{a\})$ itself can be evaluated by the same perspective formula. If $u = b(a)$, then

$$\mathbb{R}_{>0}^3 \setminus [0, u_1] \times [0, u_2] \times [0, u_3]$$

has the disjoint decomposition

$$[u_1, \infty) \times [0, \infty) \times [0, \infty) \cup [0, u_1] \times [u_2, \infty) \times [0, \infty) \cup [0, u_1] \times [0, u_2] \times [u_3, \infty).$$

Thus the absolute integral $R_2(P)$ is obtained by one singleton computation and one bounded weighted box-difference computation.

6 Weighted box integration and three-dimensional complexity

Let

$$Q = [\ell_1, u_1] \times [\ell_2, u_2] \times [\ell_3, u_3]$$

be an axis-aligned box away from the origin; infinite upper bounds are allowed. An antiderivative of $(x_1 + x_2 + x_3)^{-4}$ is

$$F(x_1, x_2, x_3) = -\frac{1}{6(x_1 + x_2 + x_3)}.$$

Therefore

$$\mu_3(Q) = \int_Q \frac{dx}{(x_1 + x_2 + x_3)^4} = \sum_{\epsilon \in \{0,1\}^3} (-1)^{3-(\epsilon_1+\epsilon_2+\epsilon_3)} F(x_1^{\epsilon_1}, x_2^{\epsilon_2}, x_3^{\epsilon_3}),$$

where $x_i^0 = \ell_i$ and $x_i^1 = u_i$. Corner terms with $u_i = \infty$ are interpreted as zero.

Assume now that a hypervolume-style routine returns a disjoint axis-aligned box decomposition of $U(b(P))$ into M boxes in time $T_{\text{decomp}}(n, M)$. Subtracting the single anchor box $U(b(a))$ from one output box creates at most a constant number of boxes in three dimensions. Each remaining box is integrated in constant time by the formula above.

Theorem 3 (Output-sensitive bound in three objectives). *Given a disjoint M -box decomposition of $U(b(P))$, the absolute integral $R_2(P)$ can be computed in*

$$T_{\text{decomp}}(n, M) + O(M)$$

time and $O(M)$ space after choosing a dominated anchor a .

Proof. Compute $R_2(\{a\})$ by the three semi-infinite complement boxes above, which is constant time. Then compute $I_a(P)$ by Theorem 2. The decomposition costs $T_{\text{decomp}}(n, M)$. For each of the M output boxes, subtracting the anchor box creates only constantly many boxes, and each weighted integral is evaluated by the eight-corner formula in constant time. Hence the additional work is $O(M)$ and the storage is linear in the emitted decomposition. $\square$

The tree-free three-dimensional hypervolume algorithm described in [2] computes the integration region by a z -sweep and array-only skyline maintenance with path-compressed skip pointers. The post states $O(n \log n)$ time and $O(n)$ memory for the three-dimensional hypervolume computation and describes the sweep decomposition of the dominated region. In the case where this routine emits $M = n$ boxes, Theorem 3 gives

$$T(n) = O(n \log n) + O(n) = O(n \log n), \quad S(n) = O(n).$$

This is the advertised three-objective complexity bound for computing the absolute integral R_2 indicator, not only the anchor-normalised improvement.

7 Generalization to N objectives

The perspective transformation is not intrinsically three-dimensional. The three-objective case is special only because it admits particularly efficient hypervolume box-decomposition algorithms. The analytic transformation, the weighted box integral, and the output-sensitive reduction all extend to arbitrary dimension N .

7.1 Correctness

For $p \in \mathbb{R}_{>0}^N$ define

$$b(p) = \left(\frac{1}{p_1}, \dots, \frac{1}{p_N} \right), \quad \text{box}(b(p)) = \prod_{i=1}^N [0, 1/p_i].$$

The N -dimensional perspective map is

$$x_i = \frac{w_i}{t}, \quad w_i = \frac{x_i}{x_1 + \dots + x_N}, \quad t = \frac{1}{x_1 + \dots + x_N}.$$

Lemma 3 (N -dimensional perspective correspondence). *For $p \in \mathbb{R}_{>0}^N$, apart from measure-zero boundaries,*

$$t > g_p(w) \iff x \in \text{box}(b(p)).$$

Moreover,

$$dw \, dt = (x_1 + \dots + x_N)^{-(N+1)} \, dx.$$

Proof. Substituting $w_i = x_i/s$ and $t = 1/s$, with $s = x_1 + \dots + x_N$, turns $t > g_P(w)$ into $1 > \max_i x_i p_i$, equivalent to $x_i < 1/p_i$ for every coordinate. For the Jacobian, use local coordinates $(w_1, \dots, w_{N-1}, t)$ with $w_N = 1 - \sum_{i=1}^{N-1} w_i$. The absolute determinant is $s^{-(N+1)}$. $\square$

Theorem 4 (Absolute N -dimensional representation). *For every nonempty $P \subset \mathbb{R}_{>0}^N$,*

$$R_2(P) = \int_{\mathbb{R}_{>0}^N \setminus U(b(P))} \frac{dx}{(x_1 + \dots + x_N)^{N+1}}.$$

Proof. As in the three-dimensional proof,

$$R_2(P) = \int_{\Delta_{N-1}} \int_0^{\tau_P(w)} dt dw.$$

The condition $t < \tau_P(w)$ is equivalent to being outside all reciprocal boxes. Lemma 3 supplies the density. $\square$

Corollary 1 (N -dimensional improvement representation). *If $a \in \mathbb{R}_{>0}^N$ is componentwise worse than all points in P , then*

$$I_a(P) = R_2(\{a\}) - R_2(P) = \int_{U(b(P)) \setminus U(b(a))} \frac{dx}{(x_1 + \dots + x_N)^{N+1}}.$$

7.2 Weighted box formula

For a box $Q = \prod_{i=1}^N [\ell_i, u_i]$ away from the origin, define $x_i^0 = \ell_i$ and $x_i^1 = u_i$. The N -dimensional weighted box integral is

$$\mu_N(Q) = \int_Q \frac{dx}{(x_1 + \dots + x_N)^{N+1}} = \frac{1}{N!} \sum_{\epsilon \in \{0,1\}^N} (-1)^{|\epsilon|} \frac{1}{x_1^{\epsilon_1} + \dots + x_N^{\epsilon_N}}.$$

Terms with an infinite coordinate are interpreted as zero.

Proof. The function

$$F_N(x) = \frac{(-1)^N}{N!(x_1 + \dots + x_N)}$$

has mixed derivative $(x_1 + \dots + x_N)^{-(N+1)}$. Applying the usual box inclusion-exclusion formula yields the expression above. $\square$

7.3 Pseudocode using a given box decomposition

Algorithm 1 assumes that another routine supplies a disjoint box decomposition of the reciprocal dominated region $U(b(P))$. This routine may be a hypervolume dimension sweep, a recursive Klee-measure routine, a quick hypervolume variant, or any other box-emitting decomposition method. The perspective algorithm consumes boxes and replaces ordinary volumes by weighted integrals.

Algorithm 1 Perspective evaluation of absolute integral R_2 in N objectives

Require: Dimension N , point set $P \subset \mathbb{R}_{>0}^N$, dominated anchor $a \in \mathbb{R}_{>0}^N$

Require: A routine BOXDECOMPOSE returning disjoint boxes for $U(b(P))$

Ensure: $R_2(P)$

```
1:  $u \leftarrow b(a)$ 
2:  $R_a \leftarrow$  weighted integral of  $\mathbb{R}_{>0}^N \setminus \text{box}(u)$  using the  $N$  semi-infinite complement boxes
3:  $A \leftarrow \text{box}(u)$ 
4:  $\mathcal{D} \leftarrow \text{BOXDECOMPOSE}(\{b(p) : p \in P\})$ 
5:  $I \leftarrow 0$ 
6: for all  $Q \in \mathcal{D}$  do
7:    $\mathcal{C} \leftarrow \text{SUBTRACTBOX}(Q, A)$ 
8:   for all  $C \in \mathcal{C}$  do
9:      $I \leftarrow I + \mu_N(C)$   $\triangleright 2^N$ -corner weighted integral
10:  end for
11: end for
12: return  $R_a - I$ 
```

7.4 Complexity

Theorem 5 (N-dimensional output-sensitive complexity). *Suppose $U(b(P))$ is decomposed into M disjoint axis-aligned boxes in time $T_{\text{decomp}}(n, M, N)$. Then Algorithm 1 computes the absolute integral $R_2(P)$ in*

$$T_{\text{decomp}}(n, M, N) + O(NM + 2^N M)$$

time and $O(M)$ space if the decomposition is stored. For fixed N , this is

$$T_{\text{decomp}}(n, M, N) + O(M).$$

Proof. The decomposition step costs $T_{\text{decomp}}(n, M, N)$. Subtracting one anchor box from one emitted box can be done by a slab decomposition with at most $2N$ pieces, so this contributes $O(NM)$ bookkeeping. Each resulting box is evaluated by summing over its 2^N corners. The N semi-infinite boxes needed for $R_2(\{a\})$ cost $O(N2^N)$, which is dominated by the main term for nontrivial M . Thus the post-processing overhead is $O(NM + 2^N M)$. For fixed dimension all factors depending only on N are constants. $\square$

Remark 1 (What generalizes). *The perspective mapping generalizes to every dimension N . The special $O(n \log n)$ bound is a three-dimensional consequence of having an $O(n \log n)$ box emitter with $M = O(n)$. In higher dimensions the resulting complexity is the complexity of the chosen N -dimensional box decomposition plus the $O(2^N M)$ weighted-integration overhead.*

8 Relation to existing work

The computation above is not an ordinary hypervolume value: the density $(x_1 + \dots + x_N)^{-(N+1)}$ is essential. What is inherited from hypervolume algorithms is the geometry and the box decomposition. Fonseca, Paquete, and López-Ibáñez [1] gave a dimension-sweep algorithm for the hypervolume indicator. The array-only implementation discussed in [2] is a tree-free version of this sweep idea in three dimensions.

For the R_2 indicator, Schäpermeier and Kerschke [3] emphasise the continuous, integrated variant and Pareto compliance. Jaszewicz and Zielniewicz [4] propose QR2, an exact method adapting quick hypervolume ideas to R_2 calculation. The present note gives a complementary route: use the perspective map to convert the Tchebycheff-shadow integral into weighted integration over a reciprocal anchored-box region. In three objectives this leads to an $O(n \log n)$ consequence whenever an $O(n \log n)$, $O(n)$ -box hypervolume emitter is available; in N objectives it gives an output-sensitive reduction to weighted box integration.

9 Reference implementation and verification

The accompanying Python script `integral_r2_perspective.py` implements the following functions:

- `perspective_r2_value(points, anchor)`, which computes the absolute integral $R_2(P)$ by reconstructing it from a dominated anchor;
- `perspective_r2_improvement(points, anchor)`, which computes $R_2(\{a\}) - R_2(P)$ from reciprocal boxes;
- `single_point_r2_perspective(anchor)`, which evaluates $R_2(\{a\})$ from the semi-infinite complement decomposition;
- `weighted_box_integral(box)`, the eight-corner formula in three objectives;
- `monte_carlo_r2_value(points)`, a scalarization-side Monte Carlo check;
- `decompose_union_boxes_sweep(corners)`, a transparent sweep-based box emitter used for verification. The weighted summation routine can consume boxes from any hypervolume decomposition routine, including an $O(n \log n)$ tree-free emitter.

The verification compares the perspective formula with a subdivision-based exact evaluator over the weight simplex and with Monte Carlo integration. The subdivision evaluator constructs the arrangement of all affine Tchebycheff pieces on Δ_2 and integrates the active lower envelope cell by cell.

Table 1: Verification of the absolute R_2 perspective implementation. Values are unnormalised simplex integrals.

instance	n	perspective R_2	subdivision R_2	abs. diff.	MC half-width
three_point_front	3	0.158359774791	0.158359774791	2.78×10^{-17}	2.03×10^{-4}
four_point_front	4	0.141937185975	0.141937185975	1.11×10^{-16}	1.72×10^{-4}
five_point_front	5	0.147175934894	0.147175934894	5.55×10^{-17}	1.85×10^{-4}

The exact perspective and subdivision values agree to floating-point precision. The Monte Carlo estimates fall within the reported sampling uncertainty. The same CSV file also reports the anchor-normalised improvement values.

10 Conclusion

The Tchebycheff shadow perspective mapping provides a direct bridge between the integral R_2 indicator and anchored-box geometry. The absolute integral $R_2(P)$ is the weighted measure of the reciprocal-space complement of the anchored-box union $U(b(P))$. For computation, a dominated anchor converts this unbounded complement integral into a singleton term minus a bounded weighted hypervolume difference. Thus the method computes the integral R_2 indicator itself, not only integral- R_2 improvement.

In three objectives, this bridge is especially useful because an $O(n \log n)$ hypervolume box emitter with $O(n)$ boxes can be combined with constant-time weighted integration per box. The resulting exact evaluation has time $O(n \log n)$ and space $O(n)$ under that decomposition assumption. The same transformation also gives a general N -objective theorem. It replaces the three-dimensional density $(x_1 + x_2 + x_3)^{-4}$ by $(x_1 + \dots + x_N)^{-(N+1)}$ and evaluates each output box by a 2^N -corner formula. Thus the general algorithm is output-sensitive: once an M -box decomposition is available, the additional weighted-integration overhead is $O(2^N M)$ for fixed anchor subtraction and fixed dimension.

A Python usage

The archive contains

```
integral_r2_perspective.py  
r2_subdivision3d.py  
verification_results.csv  
README.md
```

Run

```
python integral_r2_perspective.py
```

to regenerate `verification_results.csv`. The perspective implementation uses only NumPy. The subdivision comparison uses Shapely because it performs polygon clipping on the weight simplex.

B Implementation note

The transparent sweep decomposer included in the script is designed for readability and reproducibility. The asymptotic statement in Theorem 3 separates the hypervolume-style box emitter from the weighted integration step. Therefore, replacing the bundled box emitter by an $O(n \log n)$ tree-free emitter that outputs $M = O(n)$ boxes immediately gives the claimed $O(n \log n)$ algorithm for the same weighted integral.

References

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